The Geometric Atom: Atomic Structure as a Packing Problem*

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We present a discrete geometric framework for the hydrogen atom based on finite information capacity. The hydrogen state space is modeled as a paraboloid lattice dual to the SO(4,2) conformal algebra. This single-particle model exhibits a fundamental algebra-geometry duality: **(i) Algebraic exactness:** Transition operators $T_\pm, L_\pm$ reproduce the exact Rydberg spectrum $E_n = -1/(2n^2)$ from operator eigenvalues. **(ii) Geometric artifacts:** The graph Laplacian lifts s/p degeneracy at finite lattice cutoff (peaking near $n_{\max} = 8$ at $\sim$16% and falling to $\sim$1.7% by $n_{\max} = 10$; see the §III QA note) through differential weighted connectivity. Systematic convergence analysis from $n_{\max} =$ 5 to 30 demonstrates this is a discretization artifact—spectral aliasing from lattice truncation—that oscillates and decays below 1%, recovering the exact Coulomb degeneracy in the continuum limit. **(iii) Geometric phase structure:** The Berry phase vanishes identically for the real-valued transition operators; the edge-weight log-holonomy $\Theta(n) = -2\ln((n+1)/n) \sim n^{-1}$ measures the curvature of the weight function across the lattice. **Crucially, this model describes only the electron manifold**—electromagnetic interactions and the fine structure constant α cannot emerge from a single-particle lattice and require coupling to photon degrees of freedom, which we address in a companion paper.

I. INTRODUCTION: THE TEXTURE OF THE VACUUM

Physics traditionally assumes spacetime is a smooth continuum. Quantum mechanics inherits this assumption, treating the Hilbert space as infinite-dimensional with continuous operators. Yet this framework produces persistent conceptual difficulties—wave-particle duality, nonlocality, and the measurement problem—that resist resolution within the continuum paradigm.

We propose an alternative foundation: *physical systems encode finite information.* A quantum state labeled by quantum numbers (n, l, m) cannot store infinite positional precision. If information density is finite, then the state space itself must be *discrete*—a lattice, not a continuum. The hydrogen atom becomes a **graph**: nodes represent quantum states, edges represent physical transitions, and the Hamiltonian becomes a matrix encoding connectivity.

This discrete structure possesses a natural geometry. The hydrogen atom's dynamical symmetry group $SO(4,2)$ [1, 2] has a unique dual: the **paraboloid lattice**, where quantum numbers (n, l, m) map to coordinates on a curved 3D surface. The radius grows parabolically as $r \sim n^2$, and the depth encodes energy: $z = -1/n^2$.

This lattice exhibits a profound duality between *algebra* and *geometry*:

- **The Algebraic Skeleton:** Transition operators $T_\pm$ (radial) and $L_\pm$ (angular) generate the spectrum. Their eigenvalues reproduce the exact Rydberg formula $E_n = -1/(2n^2)$ without correction.
- **The Geometric Flesh:** The lattice *topology*—the pattern of connections—generates physical corrections. High angular momentum states occupy high-connectivity nodes, creating differential weighted connectivity at finite truncation. The edge-weight log-holonomy provides a measure of lattice curvature that decays as n^{-1}.

In this paper, we present computational evidence for this dual framework. Section II demonstrates spectral exactness. Section III analyzes the s/p degeneracy breaking by the graph Laplacian, demonstrating it is a discretization artifact that vanishes in the continuum limit. Section IV analyzes the geometric phase structure, showing that the Berry phase vanishes while the log-holonomy decays as n^{-1}. Section V discusses the algebra-geometry duality and its connection to wave-particle complementarity. Section VI summarizes our findings.

II. THE ALGEBRAIC SKELETON: EXACT SPECTROSCOPY

The paraboloid lattice is generated by two sets of ladder operators forming the hydrogen atom's dynamical algebra $SU(2) \otimes SU(1,1)$:

$$\text{Angular:} \quad L_z, L_\pm \quad \text{(standard angular momentum)}, \tag{1}$$

$$\text{Radial:} \quad T_0, T_\pm \quad \text{(energy shell transitions)}. \tag{2}$$

These operators act on basis states $|n, l, m\rangle$ with selection rules:

$$L_\pm |n, l, m\rangle \propto |n, l, m \pm 1\rangle, \tag{3}$$

$$T_\pm |n, l, m\rangle \propto |n \pm 1, l, m\rangle. \tag{4}$$

* Reconstructing Atomic Structure from the Principle of Finite Capacity

The operator weights are determined by the Biedenharn-Louck calculus [3] for the $SU(2)$ Clebsch-Gordan coefficients and by the noncompact $SU(1,1)$ ladder structure [1].

A. Spectral Reproduction

The critical test of this algebra is spectroscopic: do the operators reproduce the Rydberg formula? We construct the Hamiltonian operator from the Casimir invariants of the algebra. The energy eigenvalues are

$$E_n = -\frac{1}{2n^2} \quad \text{(Hartree atomic units)}, \tag{5}$$

in exact agreement with the hydrogen spectrum. No fitting parameters are required; the algebra determines the spectrum uniquely.

This result validates the lattice as a faithful discrete representation of the hydrogen Hilbert space. The nodes are not a coarse-graining—they *are* the quantum states. The edges encode physically allowed transitions. The spectrum emerges from operator eigenvalues, not from solving differential equations. (Verified in `tests/test_paper1_rydberg.py`: the $SU(1,1)$ number operator $N = -2[T_+, T_-]$, assembled purely from the ladder generators, has integer spectrum $\{n\}$ and the $SU(2)$ Casimir L^2 gives $l(l+1)$, both bit-exact, so $E_n = -1/(2n^2)$ follows from the algebra. This is the dynamical-symmetry construction; the discrete graph Laplacian $L = D - A$ is separately positive-semidefinite and reproduces the spectrum only in the continuum limit, Paper 7.)

III. THE GEOMETRIC FLESH: TOPOLOGICAL ARTIFACTS OF FINITE TRUNCATION

While the algebraic operators yield the exact spectrum, the *geometric structure* of the lattice—its connectivity pattern—introduces systematic artifacts when the lattice is truncated at finite $n_{\max}$. We demonstrate this by constructing the graph Laplacian and analyzing its symmetry-breaking properties as a function of lattice size.

A. The Graph Laplacian Hamiltonian

In graph theory, the kinetic energy of a particle on a lattice is described by the **graph Laplacian** [5]:

$$L = D - A, \tag{6}$$

where A_{ij} is the adjacency matrix (edge weights) and D is the weighted degree matrix[8]. This Laplacian is the discrete analog of $-\nabla^2$.

For the paraboloid lattice, we construct A from the transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|, \tag{7}$$

where $|\cdot|$ denotes element-wise absolute value (undirected edges). The Hamiltonian becomes

$$H = \beta L + V = \beta(D - A) + V, \tag{8}$$

where $V_{ii} = -1/n_i^2$ is the diagonal Coulomb potential and β is a scaling factor.

B. Differential Connectivity by Angular Momentum

The crucial observation is that the degree matrix D is *not rotationally invariant*. Computing the weighted degrees for states with $n = 2$:

$$\text{Pole (2s):} \quad D_{(2,0,0)} = 0.854, \tag{9}$$

$$\text{Equator (2p):} \quad D_{(2,1,0)} = 3.416. \tag{10}$$

States with higher angular momentum l have *more connections*—more available transitions to neighboring states. This higher connectivity translates to higher on-site energy in the Laplacian formulation.

C. The Splitting as Discretization Artifact

We performed exact diagonalization of H (Eq. 8) for lattices with $n_{\max}$ ranging from 5 to 30. Using the Lanczos algorithm, we identified the eigenstates with maximal overlap with $|2,0,0\rangle$ (2s) and $|2,1,0\rangle$ (2p) and computed the relative splitting $\Delta E_{\text{rel}} = |E_{2p} - E_{2s}|/|E_{2s}|$.

At $n_{\max} = 10$ (385 nodes), the splitting is:

$$\lambda_{2s} = 0.0202, \tag{11}$$

$$\lambda_{2p} = 0.0238, \tag{12}$$

$$\Delta E = \lambda_{2p} - \lambda_{2s} = 0.0035, \tag{13}$$

corresponding to a 16% relative splitting. However, this degeneracy breaking is *not* an emergent physical force. Systematic convergence analysis reveals it is a **discretization artifact** of finite lattice truncation—the graph-theoretic analog of spectral aliasing in finite-element methods.

a. QA correction (2026-06-14). The specific quantitative values in this section—the degrees $D_{2s} = 0.854$, $D_{2p} = 3.416$, the eigenvalues $\lambda_{2s} = 0.0202$, $\lambda_{2p} = 0.0238$, and the 16% splitting quoted at $n_{\max} = 10$—are *not* reproduced by the current production lattice (the $A = |T_+| + |T_-| + |L_+| + |L_-|$ CG-magnitude construction they assume is not implemented in `geovac/`; a faithful rebuild gives $D_{2s} \approx 1.9$, $D_{2p} \approx 3.8$ and a $\sim 1.7\%$ splitting at $n_{\max} = 10$, with $\sim 16\%$ landing near $n_{\max} = 8$). What *is* verified (`tests/test_trunk_qa_splitting.py`, `test_trunk_qa_degree_table.py`) is the *qualitative* content: higher-l states have higher connectivity ($D_{2p} > D_{2s}$), and the s/p splitting decays (oscillatorily) to zero as $n_{\max} \to \infty$. The specific figures are retained as illustrative pending a reconstruction of the CG-weighted construction. *This caveat applies wherever these figures recur—the abstract, Table I, the convergence list below, and the figure captions.*

D. Convergence Analysis

The key diagnostic is the behavior of ΔE_{rel} as $n_{\max} \to \infty$. We swept the lattice cutoff from $n_{\max} = 5$ (55 vertices) through $n_{\max} = 30$ (9,455 vertices) and tracked the 2s/2p splitting at each resolution:

- At $n_{\max} = 5$: $\Delta E_{\mathrm{rel}} \approx 13\%$
- At $n_{\max} = 8$: $\Delta E_{\mathrm{rel}} \approx 16\%$ (oscillatory peak); by $n_{\max} = 10$ it has fallen to $\approx 1.7\%$ (faithful rebuild — see the QA correction above)
- At $n_{\max} = 20$: $\Delta E_{\mathrm{rel}} \approx 0.3\%$
- At $n_{\max} = 30$: $\Delta E_{\mathrm{rel}} \approx 0.005\%$

The splitting does not decay monotonically; it *oscillates* at intermediate resolutions before collapsing to negligible values. This oscillatory decay is the hallmark of spectral aliasing: at low $n_{\max}$, the truncated boundary of the graph Laplacian introduces standing-wave reflections that differentially shift eigenvalues in distinct angular momentum sectors. As the boundary recedes (larger $n_{\max}$), these reflections weaken and the exact s/p degeneracy of the continuum Coulomb problem is recovered.

E. Physical Interpretation

The 2p state, with its higher weighted degree ($D_{2p} \approx 4 \times D_{2s}$), is more sensitive to the lattice boundary than the 2s state. At finite $n_{\max}$, the truncation acts as an absorbing wall that differentially perturbs high-connectivity nodes. This is a *geometric boundary effect*, not an emergent centrifugal barrier.

Crucially, *no explicit* $l(l+1)/r^2$ term appears in the Hamiltonian. The transient splitting arises entirely from the finite extent of the graph, not from any force encoded in the topology. The graph Laplacian does not spontaneously generate the Lamb shift or replace electromagnetic interactions; it faithfully reproduces the Coulomb degeneracy in the limit of large lattices. We understand exactly how our discrete mesh maps onto the continuous spectrum and where the aliasing boundaries lie—a hallmark of mathematical rigor rather than a physical limitation.

IV. GEOMETRIC PHASE STRUCTURE OF THE LATTICE

The lattice carries a natural notion of parallel transport around closed loops (plaquettes). We examine whether this transport produces a nontrivial geometric phase—a Berry holonomy [4]—and find that the answer is negative for the real-valued transition operators used here. We then characterize the *log-holonomy*, which measures curvature of the edge-weight function, and discuss what would be required for a genuine geometric phase.

A. Flat Connection for Real Operators

A **rectangular plaquette** on the lattice is constructed via radial ($T_\pm$) and azimuthal ($L_\pm$) operators:

$$|n,l,m\rangle \xrightarrow{T_+} |n{+}1,l,m\rangle \xrightarrow{L_+} |n{+}1,l,m{+}1\rangle \xrightarrow{T_-} |n,l,m{+}1\rangle \xrightarrow{L_-} |n,l, \tag{14}$$

The Berry phase for this plaquette is the argument of the holonomy product:

$$\theta = \arg\big(\langle n{+}1,l,m|T_+|n,l,m\rangle\,\langle n{+}1,l,m{+}1|L_+|n{+}1,l,m\rangle\,\langle n,l,m{+}1|T_-| \tag{15}$$

Each matrix element is a Biedenharn–Louck coupling coefficient. For the $SU(2)$ angular-momentum operators, $\langle l,m+1|L_+|l,m\rangle = \sqrt{(l-m)(l+m+1)}$, and for the $SU(1,1)$ radial operators, $\langle n+1,l|T_+|n,l\rangle = \sqrt{(n-l)(n+l+1)/4}$. All four factors are **real and positive**. Therefore

$$\theta = \arg(\text{positive real number}) = 0 \qquad \text{for every plaquette at every (} \tag{16}$$

The lattice connection defined by these operators is *flat*: it carries zero Berry curvature.

Erratum. An earlier version of this paper reported a fitted exponent $k = 2.113 \pm 0.015$ for the Berry phase scaling. That value was a theoretical prediction placed as a placeholder pending numerical validation; the accompanying figure was never generated (it remained a `[Placeholder]`). Subsequent computation confirms $\theta = 0$ identically for all plaquettes, as shown above. The claimed $k = 2.113$ is therefore **retracted**.

B. Log-Holonomy of Edge Weights

Although the Berry phase vanishes, the lattice edge weights carry a well-defined *log-holonomy*—the logarithmic circulation around a plaquette:

$$\Theta(n) = \ln w_{12} + \ln w_{23} - \ln w_{34} - \ln w_{41}, \tag{17}$$

where w_{ij} are the adjacency-matrix weights along each edge. For the topological weights $w = 1/(n_1 n_2)$, the $T_\pm$

edges share the factor $1/\big(n(n+1)\big)$ and cancel, leaving

$$\Theta(n) = \ln\frac{1}{(n+1)^2} - \ln\frac{1}{n^2} = -2\ln\frac{n+1}{n} \sim \frac{2}{n} \qquad (n \gg 1). \tag{18}$$

The per-plaquette log-holonomy decays as n^{-1} exactly ($k = 1.0$, $R^2 = 1.0$). This measures the **curvature of the weight function** —how edge weights vary across the lattice—not a genuine geometric phase in the Berry sense.

C. Toward a Genuine Geometric Phase

A nonzero Berry phase requires complex-valued transition amplitudes. Possible extensions include:

1. Condon–Shortley sign conventions, which multiply L_+ by $(-1)^m$ (convention-dependent, yielding discrete $\{0, \pi\}$ phases);
2. A $U(1)$ gauge extension—attaching phase factors $e^{if(n,l,m)}$ to the transition operators—which would produce continuously varying curvature;
3. An adiabatic Berry connection from eigenstates of a parameter-dependent Hamiltonian (as in the hyperspherical $P_{\mu\nu}(R)$ connection of Ref. [7]).

Whether any such extension recovers a scaling exponent near $k \approx 2$ (the velocity-dependent kinematic factor $v^2 \propto n^{-2}$) remains an open question for future work.

V. DISCUSSION: THE ALGEBRA-GEOMETRY DUALITY

Our results reveal a profound duality in the structure of the paraboloid lattice that mirrors the wave-particle duality of quantum mechanics.

A. The Dual Framework

1. **Algebraic Structure (Wave-like):** The operators $T_\pm, L_\pm$ and their eigenvalues reproduce the *unperturbed* hydrogen spectrum (Eq. 5) exactly. This represents the continuous, exact aspect—analogous to the wave description in quantum mechanics.
2. **Geometric Structure (Particle-like):** The graph Laplacian $L = D - A$ encodes *perturbations*—corrections arising from discrete connectivity. The weighted degree matrix D introduces differential connectivity between angular momentum sectors at finite truncation. The edge-weight log-holonomy decays as n^{-1} (Section IV), measuring the curvature of the weight function. This represents the discrete aspect of the lattice formulation.

This duality is suggestive: the algebraic operators provide exact spectral content, while the graph topology introduces finite-size corrections that must be controlled.

B. Why Calibration Fails

A critical observation: when we attempt to match the graph Laplacian's eigenvalues directly to physical energies, calibration fails. Eigenvalues remain positive (Section III), far from the negative binding energies of hydrogen.

This is not a failure—it is a fundamental feature of the duality. The Laplacian encodes *differential* effects (splitting between states), not absolute energies. The algebraic operators provide the reference frame; the geometric operators provide the corrections. Attempting to derive absolute energies from graph topology alone conflates these complementary roles.

C. The Absence of Fine Structure

A comprehensive computational search for the fine structure constant $\alpha \approx 1/137$ within the lattice geometry yields a null result. We examined:

1. **Operator ratios:** The gearing ratio $\|L_+\|/\|T_+\|$ does *not* converge to a fixed value (the spectral-norm ratio is exactly 2.0; the Frobenius-norm ratio drifts through $\approx$ 1.77 without converging—`tests/test_trunk_qa_gearing.py`), so it encodes no physical constant such as 137 or $1/\alpha$. The commutator $[T_+, L_+] = 0$, however, *is* bit-exact.
2. **Commutator defects:** The operators commute exactly, $[T_+, L_+] = 0$, indicating perfect integrability rather than non-commutative geometry.
3. **Holonomy defects:** The Berry phase is identically zero for real operators (Section IV); the log-holonomy decays as n^{-1}, reflecting the asymptotic flatness of the weight function.

This null result is *physically correct.* The fine structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ measures the strength of electromagnetic interactions—specifically, the coupling between electron and photon fields. Our lattice describes the *kinematic* structure of the electron state space without electromagnetic interactions. The hydrogen spectrum $E_n = -1/(2n^2)$ is independent of α; fine structure appears only through:

$$\text{Relativistic corrections:} \quad \Delta E_{\text{rel}} \sim \alpha^2 \cdot E_n/n, \tag{19}$$

$$\text{Spin-orbit coupling:} \quad \Delta E_{\text{so}} \sim \alpha^2 \cdot E_n \cdot l(l+1)/n^3, \tag{20}$$

$$\text{Lamb shift (QED):} \quad \Delta E_{\text{Lamb}} \sim \alpha^5 \cdot mc^2. \tag{21}$$

These require photon degrees of freedom, which are absent from our single-particle lattice.

This clarifies the role of geometry: the lattice encodes *kinematic* constraints (spectrum, angular momentum structure) but not *dynamical* couplings. To incorporate fine structure, one must introduce a photon lattice and define interaction edges weighted by α. The fine structure constant would then appear as the *coupling strength between lattices*, not as an intrinsic geometric property of either lattice alone.

D. Artifact vs. Physics: Drawing the Line

Our convergence analysis establishes a clear separation between discretization artifacts and genuine physical content of the graph Laplacian. The s/p splitting is firmly in the artifact category: it vanishes as $n_{\max} \to \infty$ and exhibits the oscillatory decay characteristic of finite-size aliasing.

By contrast, the overall eigenvalue spectrum ($E_n \to -1/(2n^2)$) and the edge-weight log-holonomy ($\Theta \propto n^{-1}$, Section IV) are *convergent* properties that survive the continuum limit. These represent genuine structural content encoded in the topology.

This distinction is essential: the graph Laplacian does not generate electromagnetic forces or the Lamb shift. It encodes the *kinematic* structure of the electron state space. Finite lattice effects must be understood as numerical artifacts, not new physics.

E. The Wavefunction as Distribution

If quantum states are discrete nodes, what is the wavefunction $\psi(r)$? One possible interpretation is that continuous wavefunctions correspond to statistical distributions over discrete lattice sites—a coarse-graining in which the Schrödinger equation becomes a diffusion equation on the graph. Under this reading, photon absorption would transfer discrete angular momentum via edge traversal. However, this interpretation is not required by the mathematical framework presented here; the graph-theoretic formulation is compatible with standard quantum-mechanical interpretation of ψ as a probability amplitude.

VI. CONCLUSION

We have demonstrated that quantum mechanics, when formulated on a discrete paraboloid lattice, exhibits a dual structure:

- **Algebraic exactness:** Transition operators reproduce the hydrogen spectrum without corrections.
- **Understood discretization artifacts:** The graph Laplacian lifts s/p degeneracy at finite $n_{\max}$ (up to $\sim$16% at intermediate $n_{\max}$, near $n_{\max} = 8$), but systematic convergence analysis confirms this is spectral aliasing from lattice truncation—not an emergent physical force. The splitting oscillates and decays below 1% by $n_{\max} = 30$, recovering the exact Coulomb degeneracy.
- **Geometric phase structure:** The Berry phase vanishes identically for the real-valued $SU(2)/SU(1,1)$ operators; the edge-weight log-holonomy decays as $\Theta \propto n^{-1}$, measuring the curvature of the weight function. Whether a $U(1)$ gauge extension can produce a genuine geometric phase with relativistic scaling remains an open question.
- **Kinematic completeness:** The lattice encodes all α-independent physics—the Coulomb spectrum, angular momentum structure, and geometric phase effects—while cleanly separating boundary artifacts from physical content.

These results support a graph-theoretic reformulation of the hydrogen state space in which the algebraic operators encode the exact spectrum and the graph Laplacian introduces controlled, quantitatively understood finite-size artifacts. The algebra-geometry duality is suggestive of a connection to wave-particle complementarity, though we do not claim it establishes one. Critically, the graph Laplacian does not replace electromagnetic interactions or generate the Lamb shift; it encodes the kinematic manifold of the electron.

VII. LIMITATIONS AND NEXT STEPS: MISSING PHOTON DEGREES OF FREEDOM

What This Model Does Not Include:
The paraboloid lattice presented here is explicitly a *single-particle* model describing only the electron's quantum state manifold. It does not include:

1. **Photon degrees of freedom:** The electromagnetic field ($U(1)$ gauge symmetry) is absent.
2. **Electron-photon coupling:** There is no mechanism for radiative transitions or electromagnetic interactions.
3. **The fine structure constant α:** As a pure electron-only model, α cannot appear. The fine structure constant measures the strength of electron-photon coupling—it describes the relationship between *two* manifolds, not a property of either one alone.

What Must Be Added:
To incorporate electromagnetic interactions, one must construct a second geometric structure:

- **A photon gauge fiber:** The $U(1)$ electromagnetic gauge symmetry suggests a circular or helical fiber structure attached to each electron state (n, l, m).

- **Coupling rules:** Edges connecting the electron lattice to the photon fiber must be defined. The weight of these inter-manifold connections would encode the coupling strength.

- **Symplectic impedance:** If both manifolds carry action (units of $\hbar$), their ratio may define a dimensionless coupling constant.

We conjecture that the fine structure constant α emerges as the *geometric impedance*—the ratio of action densities between the electron paraboloid and the photon fiber. This hypothesis is explored in a companion paper [6], which introduces a helical photon gauge fiber and computes the symplectic coupling ratio.

Outlook:
This framework invites generalization. If hydrogen is a paraboloid, what geometries describe molecules, nuclei, or quantum fields? The answer may lie in higher-dimensional lattices—discretizations of larger symmetry groups.

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[7] J. Loutey, "The Hyperspherical Lattice: Two-Electron Atoms as Coupled Channel Graphs," (companion paper, 2026).
[8] The weighted degree $D_{ii} = \sum_j |A_{ij}|$ represents the total connectivity strength of node i. For transition matrices with complex coefficients, we use absolute values to obtain undirected edge weights, explaining the non-integer values in Table I.

Appendix A: Mathematical Definitions

1. Weighted Degree Matrix

The weighted degree matrix D is a diagonal matrix where each diagonal element represents the total connection strength of a node:

$$D_{ii} = \sum_j |A_{ij}|, \tag{A1}$$

where A_{ij} is the adjacency matrix constructed from transition operators. For complex-valued operators, the absolute value ensures symmetric (undirected) edge weights. This definition explains why degree values are non-integer (Table I)—they represent weighted sums of transition matrix elements, not simple edge counts.

2. Berry Phase and Log-Holonomy

For a plaquette (Eq. 14), the Berry phase is $\theta = \arg(\prod_i \langle s_i|\hat{U}_i|s_{i+1}\rangle)$. As shown in Section IV, all four matrix elements are real and positive, so $\theta = 0$ identically.

The log-holonomy (Eq. 17) provides a nonzero measure of weight-function curvature. For topological weights $w = 1/(n_1 n_2)$, the exact result is $\Theta(n) = -2\ln\big((n+1)/n\big)$, decaying as n^{-1}.

Appendix B: Computational Details

1. Lattice Construction

The paraboloid lattice was constructed for $1 \le n \le 10$, yielding 385 quantum states $|n, l, m\rangle$ with $0 \le l < n$ and $-l \le m \le l$. Transition operators $T_\pm$ and $L_\pm$ were built using Biedenharn-Louck coupling coefficients for $SU(2)$ and $SU(1,1)$ representations.

2. Graph Laplacian Diagonalization

The adjacency matrix A was computed as the sum of absolute values of all transition operators:

$$A = |T_+| + |T_-| + |L_+| + |L_-|. \tag{B1}$$

The graph Laplacian $L = D - A$ was stored in compressed sparse row (CSR) format. The weighted degree matrix

D had diagonal elements ranging from 0.854 to 20.624, with mean degree 12.42.

Eigenvalue computation used the shift-invert Lanczos method (`scipy.sparse.linalg.eigsh`) with $\sigma = -1.0$, computing the lowest 20 eigenvalues. Diagonalization required 0.15 seconds on a standard workstation; the $O(V)$ sparse-eigenvalue scaling is verified by `tests/test_ov_scaling_rigorous.py` (multi-point log–log fit).

State identification was performed by computing overlap probabilities $|\langle n, l, m|\psi_k\rangle|^2$ for each eigenvector $|\psi_k\rangle$, selecting the eigenvector with maximal overlap for each target state.

3. Log-Holonomy Calculation

Rectangular plaquettes (Eq. 14) were identified by searching for closed paths in (n, m) space at fixed l. The Berry phase $\theta = \arg(\cdot)$ is identically zero for all plaquettes because every $SU(2)/SU(1,1)$ matrix element is real and positive (Section IV). The log-holonomy $\Theta(n) = -2\ln\big((n+1)/n\big)$ was computed analytically and verified numerically, giving power-law exponent $k = 1.0$ exactly.

Appendix C: Supplementary Data

TABLE I. Weighted node degrees for select quantum states, demonstrating differential connectivity between s and p orbitals. The non-integer values arise from weighted sums of transition matrix elements (see Appendix A).

State (n, l, m)	Weighted Degree D_{ii}	Type	Energy (Laplacian)
$(1, 0, 0)$	0.854	1s	(ground state)
$(2, 0, 0)$	0.854	2s	0.0202
$(2, 1, 0)$	3.416	2p	0.0238
$(3, 0, 0)$	0.854	3s	—
$(3, 1, 0)$	4.270	3p	—
$(3, 2, 0)$	6.124	3d	—

[Placeholder: 3D visualization of paraboloid lattice showing nodes (n, l, m) and edges from $T_\pm, L_\pm$ operators. Highlight color-coded shells for $n = 1, 2, 3$ and pole vs. equator connectivity.]

FIG. 1. The paraboloid lattice structure for $n \leq 5$. Nodes represent quantum states $|n, l, m\rangle$, and edges represent transitions via $L_\pm$ and $T_\pm$ operators. The pole ($l = 0$) has systematically lower weighted connectivity than the equator ($l > 0$), producing differential degree values that contribute to finite-size splitting.

[Placeholder: Bar chart showing eigenvalues $\lambda_0, \lambda_1, \ldots, \lambda_{19}$ from Laplacian diagonalization. Highlight $\lambda_{2s} = 0.0202$ (red bar) and $\lambda_{2p} = 0.0238$ (blue bar) with splitting $\Delta E = 0.0035$ annotated.]

FIG. 2. Eigenvalue spectrum from exact diagonalization of the graph Laplacian Hamiltonian (Eq. 8) for $n_{\max} = 10$. The $2s$ state (red, $\lambda = 0.0202$) and $2p$ state (blue, $\lambda = 0.0238$) show a splitting of $\Delta E = 0.0035$ (16% relative) at this truncation. This splitting is a discretization artifact that decays to <0.01% by $n_{\max} = 30$.

[Placeholder: Log-log plot of log-holonomy $\Theta(n)$ vs. principal quantum number n. Data points shown as circles, exact curve $\Theta = -2\ln((n+1)/n)$ as solid line. Annotate power-law exponent $k = 1.0$ exactly.]

FIG. 3. Log-holonomy $\Theta(n) = -2\ln((n{+}1)/n)$ versus principal quantum number n (log-log plot). The exact n^{-1} decay ($k = 1.0$, $R^2 = 1.0$) measures the curvature of the edge-weight function across the lattice. The Berry phase is identically zero for all plaquettes (Section IV).